

★ WEEK 5 · PART 1 · PRACTICE

The ARCH Model, Worked by Hand

Turn One Shock Into Tomorrow's Risk, Then Test For It

The ARCH(1) recursion, stationarity, and the LM test

Part 1's core idea, with the arithmetic shown. We run the ARCH(1) recursion so a big shock spikes the next day's variance and then decays - volatility clustering - then check stationarity through the unconditional variance and test whether ARCH effects are there to model at all.

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PART 1 · APPLICATIONS

When Yesterday's Shock Sets Today's Risk

Case A: run the ARCH(1) recursion and watch a big shock spike tomorrow's variance, then fade - the clustering that constant-variance models miss. Case B: check stationarity through the unconditional variance and test for ARCH effects with the LM test. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

Variance is conditional, not constant

Markets aren't equally risky every day. ARCH lets today's variance depend on yesterday's shock, so risk clusters.

2

One squared shock drives the next variance

The ARCH(1) recursion adds a fraction of yesterday's squared shock to a baseline - a big move today means a risky tomorrow.

3

Check stationarity, and test first

The unconditional variance exists only if the coefficient is below one, and an LM test confirms there are ARCH effects to model.

Everything here is a calculator exercise: one recursion, one unconditional variance, one test.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the ARCH parameters and shocks.
- Run the recursion, or test and check stability.
- Then read the risk.

What you should watch

- How a big shock spikes the next variance.
- How the spike decays back toward the mean.
- How the LM test flags clustering.

The ARCH(1) Recursion

One shock, turbulent market
A calm market can turn turbulent overnight after a big move.

ARCH(1) captures that: today's variance is a baseline plus a fraction of yesterday's squared shock. We run the recursion through a sequence that includes a crash-sized shock, and watch the conditional variance spike, then decay back toward its long-run

The model

- Baseline omega of 0.5.
- Coefficient alpha of 0.6 on the last shock.
- Return equals volatility times a normal draw.

What we'll do by hand

- Apply the recursion to a calm shock.
- Apply it to a big shock.
- Watch the spike decay.

The ARCH(1) Model

Variance from yesterday's squared shock

$$r_t = \sigma_t z_t, \quad z_t \sim N(0, 1);$$

$$\underbrace{\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2}_{\text{today's variance from yesterday's shock}}$$

today's variance from yesterday's shock

Size from direction

- Every return is volatility times a direction.
- Squaring the shock strips the sign.
- Only the magnitude drives risk.

The clustering link

- A big squared shock lifts today's variance.
- So turbulence follows turbulence.
- That single link creates clustering.

Calm, Then a Crash

The recursion on two shocks

$$\begin{aligned} \text{calm shock } 0.5 : \sigma^2 &= 0.5 + 0.6(0.25) = 0.65 \text{ (vol } 0.81) \\ \text{big shock } -3.0 : \sigma^2 &= 0.5 + 0.6(9.0) = \mathbf{5.90} \text{ (vol } \mathbf{2.43}) \end{aligned}$$

A calm shock of 0.5 leaves the variance near baseline at 0.65. But a crash-sized shock of -3.0 sends it to 5.90 - a volatility of 2.43%, more than double the long-run level. One big move rewrites tomorrow's risk.

The Spike Decays

Back toward the long-run level

next day, shock 0.8 : $\sigma^2 = 0.5 + 0.6(0.64) = 0.88 \Rightarrow$ the spike decays back toward 1.25

What happens next

- The day after the crash, a normal shock returns.
- The variance falls to 0.88.
- It's already settling down.

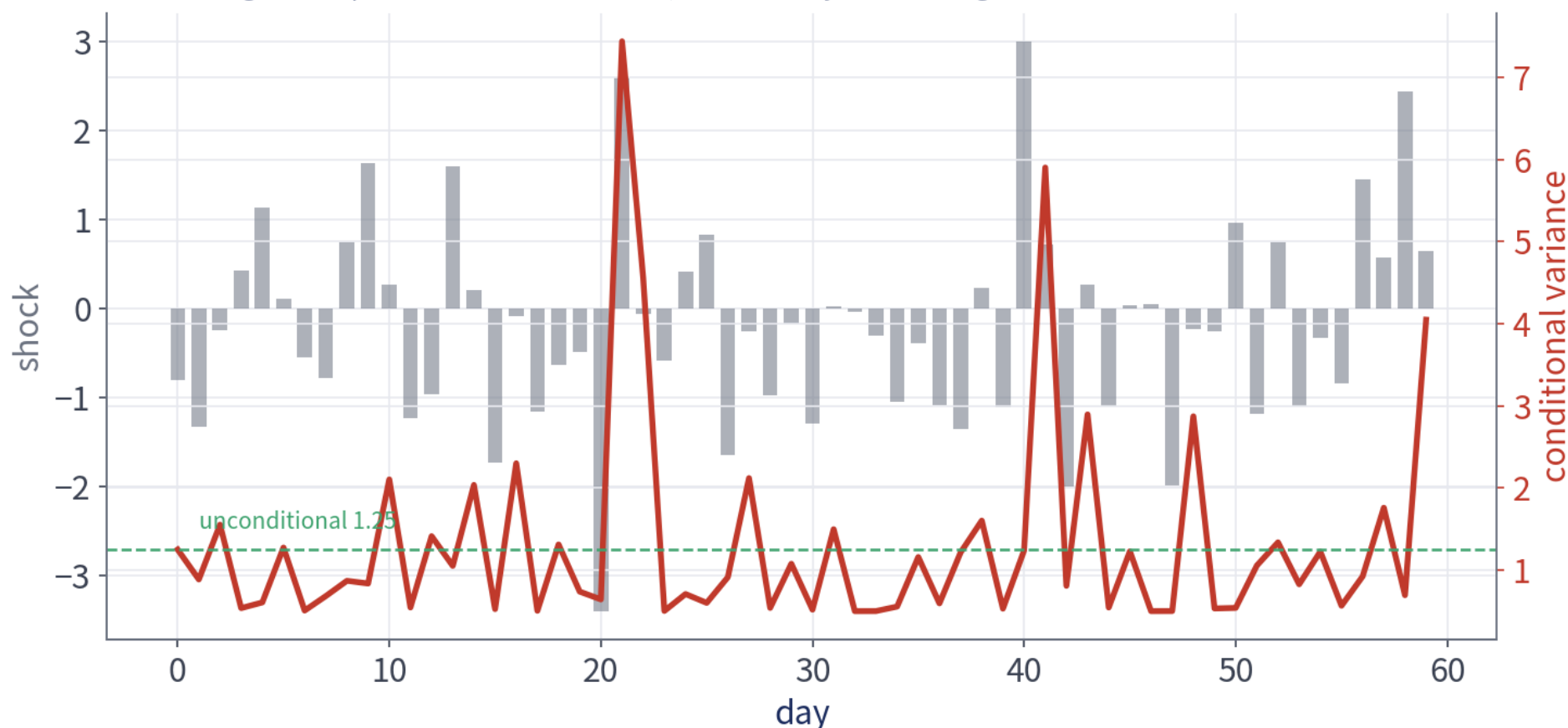
The pattern

- A spike, then a gradual return.
- Toward the unconditional 1.25.
- Clusters form, then dissolve.

Turbulence Follows Turbulence

The grey bars are daily shocks; the red line is the conditional variance. Each large shock lifts the variance sharply, and it decays back toward the dashed long-run level of 1.25 - until the next big move. Calm and stormy periods cluster exactly as real markets do.

Case A: a big shock spikes tomorrow's variance, then it decays - clustering



Why Clustering Matters

Risk is not the same every day

What constant variance misses

- It treats every day as equally risky.
- So it underestimates risk after a shock.
- And overestimates it in calm spells.

What ARCH captures

- Risk that rises right after a big move.
- A time-varying variance you can forecast.
- The clustering markets actually show.

Well-Behaved, and Worth Modeling

Stationarity, then the ARCH test

Before trusting an ARCH model, two checks. First, is it stationary - does a long-run variance even exist? That requires the coefficient below one. Second, are there ARCH effects to model at all? A quick LM test on squared residuals answers that. We compute the unconditional variance and run the test by hand

The two checks

- Stationarity: coefficient below one.
- ARCH effects: an LM test on squared residuals.
- Both come before estimation.

What we'll do by hand

- Compute the unconditional variance.
- See what happens if the coefficient exceeds one.
- Run the LM test statistic.

Unconditional Variance

The long-run level, and when it exists

$$\bar{\sigma}^2 = \frac{\omega}{1-\alpha} = \frac{0.5}{1-0.6} = \mathbf{1.25} \quad (\text{stationary iff } \alpha < 1)$$

The formula

- Baseline over one minus the coefficient.
- It's the level the variance reverts to.
- The dashed line in the last chart.

The condition

- It only makes sense if the coefficient is below one.
- At one or above, the variance has no finite mean.
- The model would explode.

Stationary or Explosive

The coefficient decides

Stationary: $\alpha = 0.6$

- Unconditional variance 0.5 over 0.4.
- Equals 1.25 - finite and well-behaved.
- Shocks fade, variance mean-reverts.

Explosive: $\alpha = 1.2$

- 0.5 over one minus 1.2 is negative.
- No finite long-run variance exists.
- Each shock amplifies - the model blows up.

Stationarity is the multivariate-free version of the unit-root condition: the coefficient must stay below one for the variance to settle rather than explode.

Testing for ARCH Effects

Is there volatility to model?

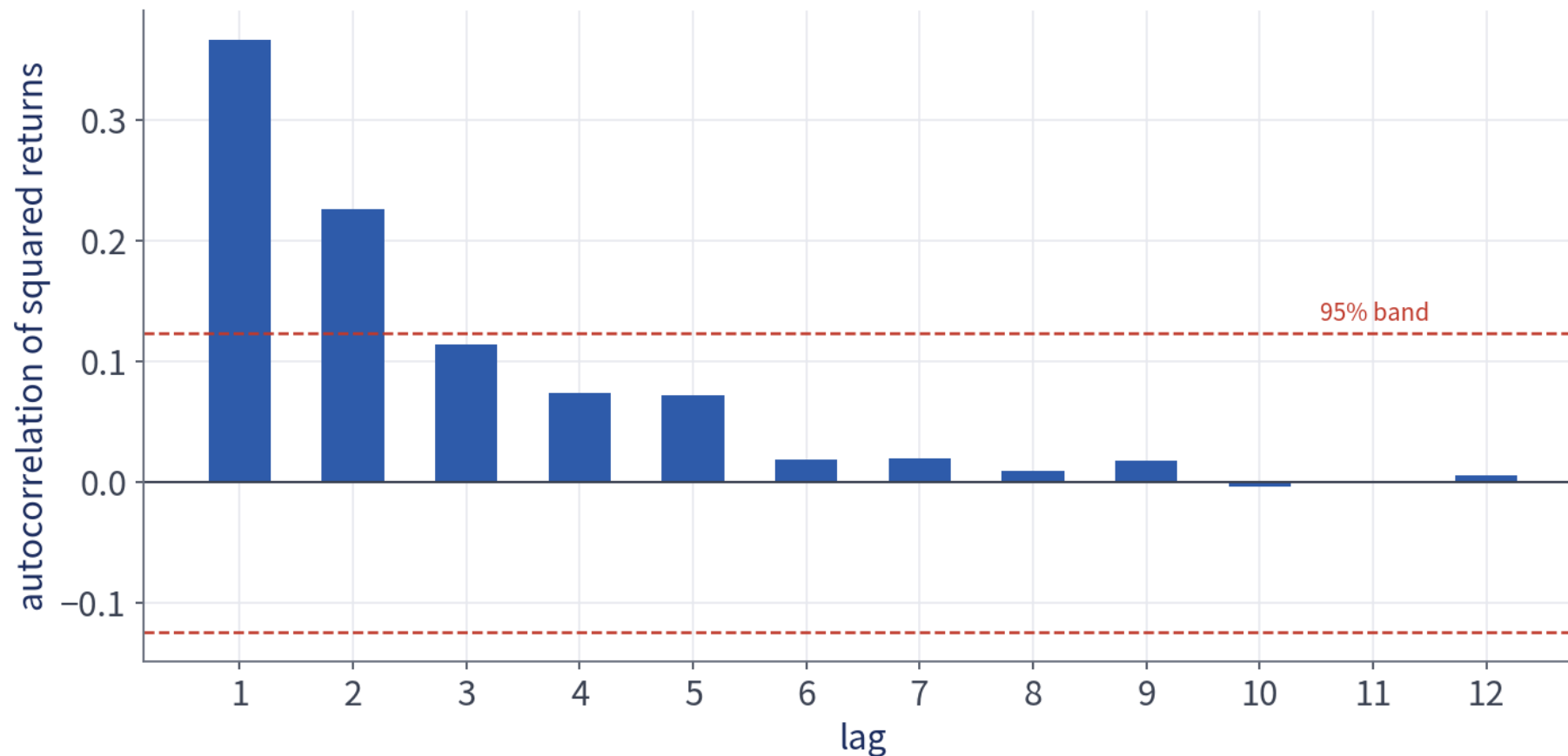
$$LM = T \cdot R^2 = 250 \times 0.08 = \mathbf{20.0} > \underbrace{3.84}_{\chi^2_{1} \ 5\%} \Rightarrow \text{ARCH effects present}$$

Regress the squared residuals on their lag and scale the fit: the LM statistic is the sample size times R-squared, here 250 times 0.08, or 20.0. That far exceeds the chi-square cutoff of 3.84 - ARCH effects are present.

The Signature of ARCH

Returns themselves show little autocorrelation, but their squares do - a slowly decaying set of positive bars that pierce the significance band. That persistence in squared returns is exactly what the LM test detects, and what tells you there is volatility clustering to model

Case B: squared returns are autocorrelated - the signature of ARCH effects



Check Before You Model

Two guards on the ARCH fit

Stationarity guards the forecast

- A coefficient below one keeps variance finite.
- The forecast reverts to a stable level.
- Above one, the model is useless.

The test guards your effort

- No ARCH effects means nothing to model.
- The LM test confirms clustering is real.
- Only then is an ARCH model worth fitting.

The ARCH Routine

Test, estimate, check, forecast.

The workflow is short. Test the squared residuals for ARCH effects; if they're there, estimate the recursion; confirm the coefficient is below one for stationarity; then forecast the variance forward. The one link - today's variance from yesterday's squared shock - turns isolated shocks into the clustered volatility markets actually

The recursion (Case A)

- Calm shock: variance 0.65.
- Crash shock: variance 5.90.
- Then it decays toward 1.25.

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The checks (Case B)

- Stationary at alpha 0.6 (variance 1.25).
- LM test 20.0 beats 3.84.
- ARCH effects confirmed.

Reproduce the Recursion and the Test

Direct Claude Code to check your arithmetic

GOAL Run the ARCH(1) recursion, check stationarity, and test for ARCH effects.

[INPUT] Prompt · Claude Code

```
Write and run part1_practice.py that, with
omega=0.5, alpha=0.6:
CASE A - recursion:
  * for shocks [0.5, -3.0, 0.8, -0.4], print each
    conditional variance and its volatility.
CASE B - checks:
  * print the unconditional variance;
  * show alpha=1.2 gives no finite variance;
  * with T=250, R^2=0.08, print the LM statistic
    and compare to 3.84.
Then plot the shocks and the conditional variance.
```

Check against the slides

- calm 0.65, crash 5.90.
- unconditional variance 1.25.
- alpha=1.2 explodes.
- $LM = 20.0 > 3.84$.

Deliverable: part1_practice.py whose printed numbers match the recursion and the tests.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What drives the ARCH(1) conditional variance?

A fraction of yesterday's squared shock, plus a baseline

Q2 When is an ARCH(1) model stationary?

When the coefficient alpha is below one

Q3 How do you test for ARCH effects?

An LM test: T times R-squared on squared residuals

All three come from the recursion, the unconditional variance, and the LM test.

Three Pitfalls to Avoid

Where ARCH trips people

Common mistakes

- "Volatility is roughly constant day to day."
- "Fit ARCH without checking the coefficient."
- "Model volatility without testing for it."

The reality

- Risk clusters - it spikes after big moves.
- A coefficient at or above one makes it explode.
- Run the LM test first; there may be nothing to model.

Two Calculations to Remember

Part 1 applications in one screen

1

One squared shock rewrites tomorrow's risk

The ARCH(1) recursion took a calm shock of 0.5 to a variance of 0.65, but a crash-sized -3.0 to 5.90 - a 2.43% volatility, more than double the long-run level - which then decayed back. That is volatility clustering, made concrete.

2

Check stationarity, and test for effects

With a coefficient of 0.6, the unconditional variance is a finite 1.25; at 1.2 it would explode. And an LM statistic of 20.0 - the sample size times R-squared - far exceeded the 3.84 cutoff, confirming ARCH effects are real.

3

A short, disciplined routine

Test the squared residuals, estimate the recursion, confirm stationarity, then forecast the variance. One link between shock and next-day risk is all it takes to model clustering.

Next — Part 2 practice: GARCH, which adds a memory of yesterday's variance for far more parsimonious volatility.